

# MedVAE: adaptation, fine-tuning and analysis for medical imaging and coronary segmentation

Yanic Rothlingshofer   Théo Palagi   Elias Corlou   Théophile Nadiedjoa

*Télécom Paris*

Supervisor: Elsa Angelini



**Context.** Deep learning in medical imaging is limited by the high resolution required for diagnostic detail. VAEs compress the image into a compact latent.

- **MedVAE:** generic medical compressor ( $> 1\text{M}$  images),  $\times 4$  per side,  $\times 16$  fewer values ( $512^2 \rightarrow 128^2 \times 1$ ), trained in two stages (reconstruction then semantic enrichment).

**Domain.** Coronary angiography (ARCADE, 1 500 images): thin, tubular, low-contrast structures, noise and artefacts.

## Central question

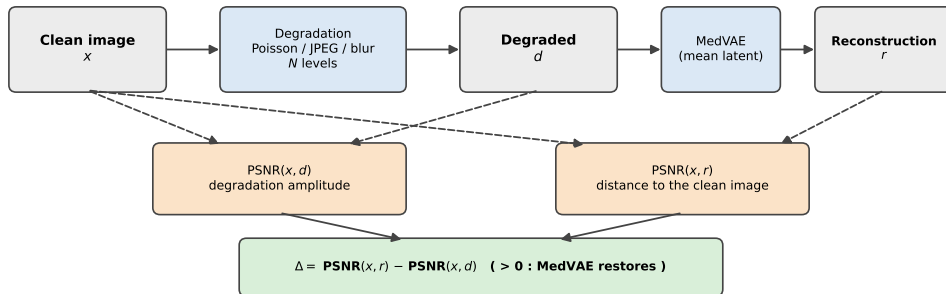
Does a generic medical compressor remain **reliable and useful** when facing a specific and demanding clinical domain?

### Four complementary axes:

- 1 Robustness to degradations
- 2 Image-quality inductive bias via FiLM
- 3 Semantic structuring of the latent (JEPA)
- 4 Latent representation for segmentation

## Robustness of MedVAE to input degradations

# Method: evaluation protocol



The reconstruction is compared to the **clean** image: does MedVAE bring a degraded input closer to it? 100 seg\_val images, deterministic latent, fixed intensity scale, seeded noise, bootstrap CIs.

# Why not a no-reference metric?

Discarded approach: score the input quality with ARNIQA (no-reference), then correlate it with the reconstruction quality.

---

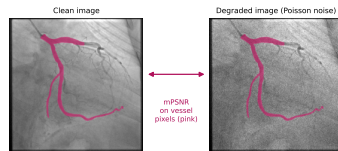
## ARNIQA — clean dataset (n=1000)

---

|                   |                          |
|-------------------|--------------------------|
| min / max         | 0.2695 / 0.6512          |
| mean $\pm$ std    | 0.4570 $\pm$ 0.0482      |
| quantiles 5/50/95 | 0.3772 / 0.4566 / 0.5391 |

---

**Weak correlation with the reconstruction PSNR** (ARNIQA:  $r = -0.37$ , wrong sign; custom score:  $r \approx -0.01$ ): the quality variance between clean images is too small to serve as an exploitable axis.



mPSNR illustration (vessel mask)

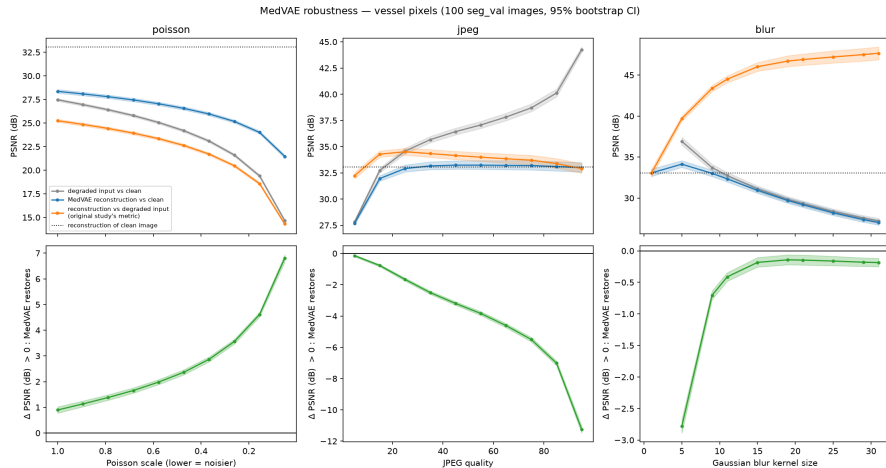
# Result: MedVAE reduces noise (less than a simple filter), not blur or JPEG

| Degradation               | $\Delta$ PSNR (dB)       | Restored |
|---------------------------|--------------------------|----------|
| Poisson                   | +1.2 $\rightarrow$ +6.6  | 99–100%  |
| $5 \times 5$ filter       | +6.3 $\rightarrow$ +10.3 | 100%     |
| JPEG (95 $\rightarrow$ 5) | −11.7 $\rightarrow$ −0.5 | 0–9%     |
| Blur                      | −2.0 $\rightarrow$ +0.1  | –        |

$$\Delta = \text{PSNR}(\text{clean}, \text{recon}) - \text{PSNR}(\text{clean}, \text{degraded})$$

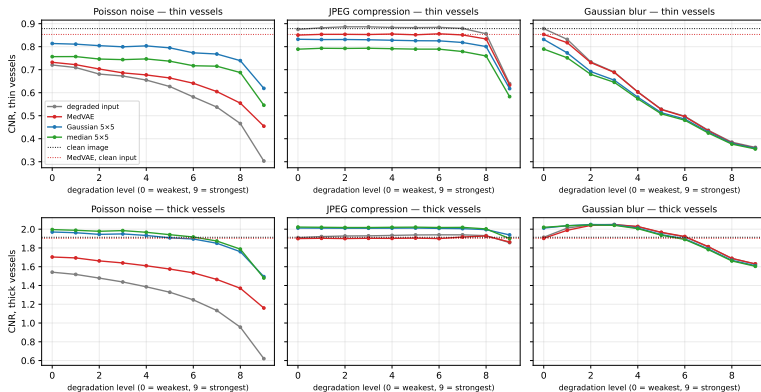
- **Poisson**: noise reduced, but 3.7–5.6 dB less than a simple  $5 \times 5$  Gaussian filter
- **JPEG**: never restored — MedVAE’s own error (33 dB ceiling) exceeds the artefacts
- **Blur**: not restored — the reconstruction follows the blurred input
- The earlier “blur reconstructs better” measured fidelity to the *degraded* input (orange)

# Result: restoration curves (vessel pixels)



Grey: degraded input vs clean. Blue: reconstruction vs clean. Orange: reconstruction vs degraded input (earlier metric). Bottom:  $\Delta$ .

# Vessel visibility: contrast-to-noise ratio (CNR)

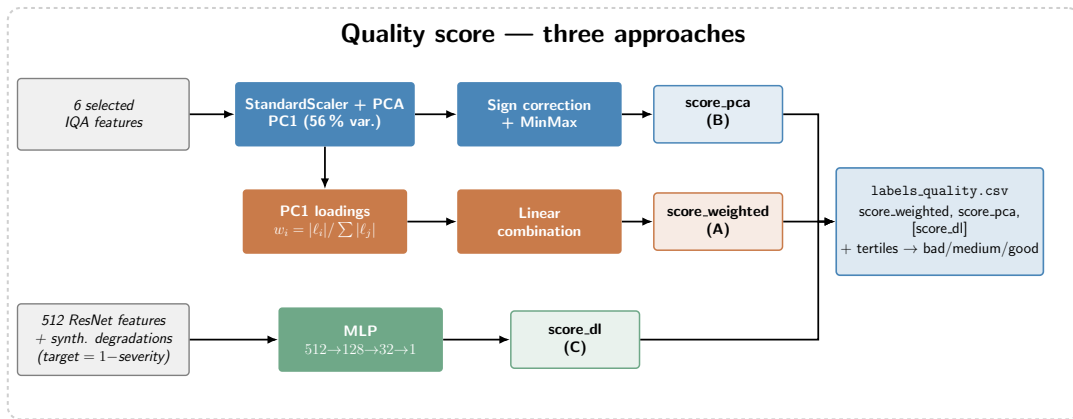


- $CNR = (\mu_{bg} - \mu_{vessel}) / \sigma_{bg}$ , thin ( $\leq 6$  px) and thick ( $\geq 12$  px) vessels
- **Poisson:** MedVAE raises the CNR (thick  $+0.16 \rightarrow +0.54$ , thin  $+0.01 \rightarrow +0.15$ ), but the  $5 \times 5$  filter does better, **thin vessels included**
- Without noise (JPEG, blur): MedVAE leaves the CNR unchanged; next step: detectability  $d'$  of a model observer

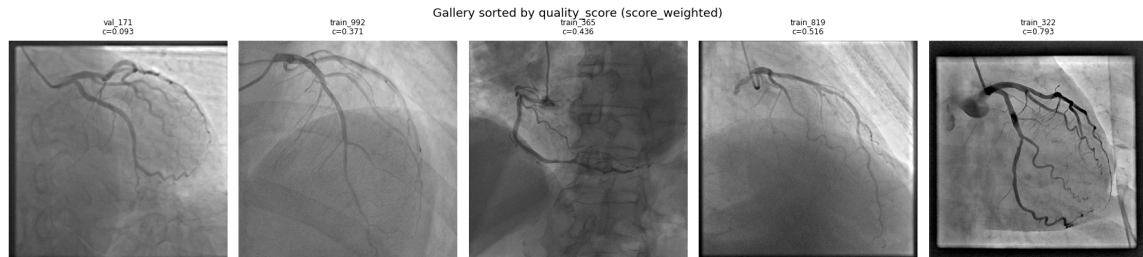
# Image-quality inductive bias (FiLM)

- MedVAE applies the **same transform** to a clean or a heavily degraded coronary angiography: no notion of *how degraded* the input is.
- **Hypothesis:** conditioning MedVAE on a quality score  $c \in [0, 1]$ , injected via FiLM, should let it adapt to the degradation level actually present.
- Three independent ways to compute  $c$  (A: weighted, B: PCA, C: learned), each with 3 seeds.

# Building the quality score $c$

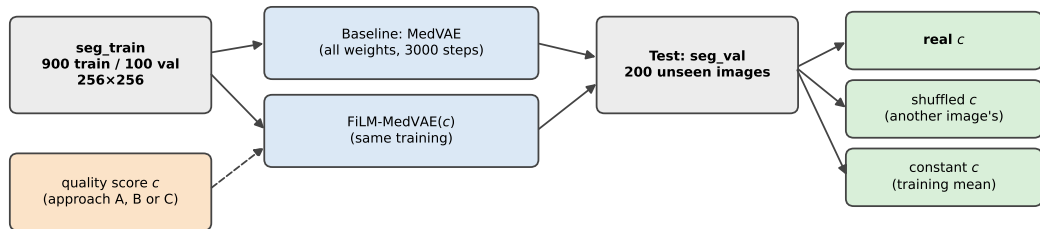


# Does the score make visual sense?



ARCADE images sorted by increasing score  $c$  (approach A).

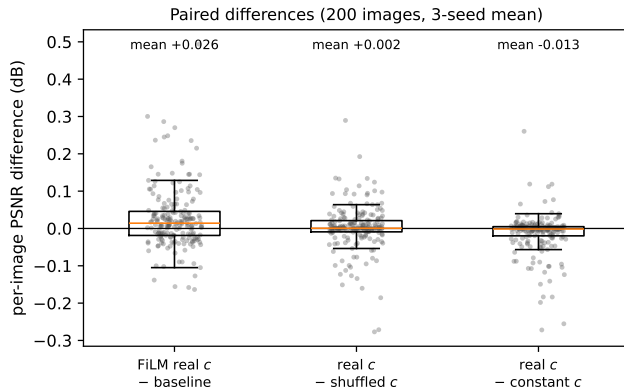
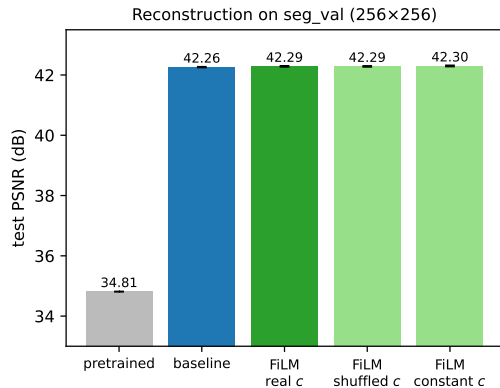
# FiLM conditioning & revised protocol



*PSNR, vessel PSNR, SSIM, HaarPSI · 3 seeds · paired per-image differences  
If the model uses c: real c > shuffled c and real c > constant c*

- FiLM after each of the 19 ResNet blocks:  $h \leftarrow (1 + \gamma(c)) h + \beta(c)$ , zero-initialised (+1.7M parameters, +3 %)
- **Revised:**  $256 \times 256$  (was  $64 \times 64$ , where MedVAE only reaches 20.7 dB), KL normalised per pixel, same training for both models, 3 seeds

# Result: fine-tuning helps, conditioning does not



- Fine-tuning MedVAE on ARCADE: **+7.4 dB** (vessels: +5.3 dB), same for all seeds
- FiLM vs baseline: +0.03 dB — negligible

# Does the network use $c$ ?

| PSNR (dB), 3 seeds | true — shuffled $c$ | FiLM — base |
|--------------------|---------------------|-------------|
| Score C            | +0.002              | +0.03       |
| Score A            | +0.007*             | +0.04       |
| Score B            | +0.033*             | +0.05       |

\* CI excluding 0, same sign for every seed; FiLM — baseline: CI over seeds includes 0

- **Score C: no** — it barely varies in training (79 % above 0.9)
- **Scores A, B: yes, but barely** ( $< 0.04$  dB, max +0.033)
- FiLM gains come from the added parameters, not from the quality information
- $c$  is computed *from the image*: for an autoencoder of that image, it carries nothing new
- The earlier “only C is exploited” compared two separately trained single runs

## Semantic structuring of the latent space (JEPA)

# Original stage 2 — BioMedCLIP distillation

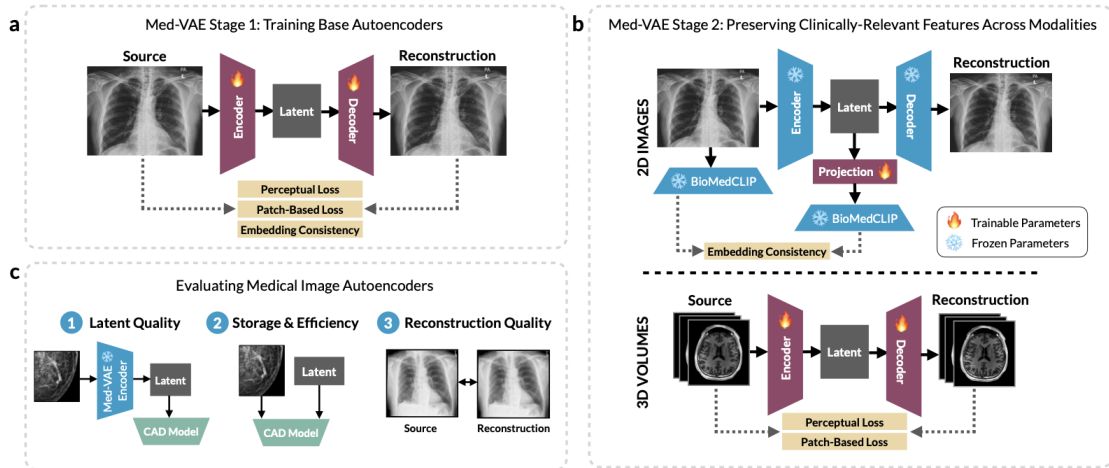
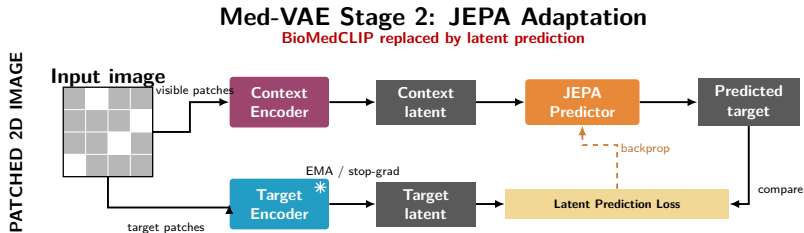


Figure reproduced from Varma et al., MedVAE (2025), Fig. 1.

# Modified stage 2 — JEPA adaptation



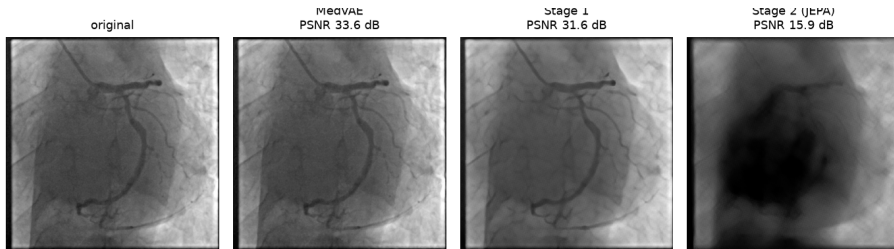
$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{SIGReg}} \mathcal{L}_{\text{SIGReg}}$$

$$\mathcal{L}_{\text{pred}} = \frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} \text{SmoothL}_1 \left( \hat{z}_t^{(p)}, \tilde{z}_t^{(p)} \right), \quad \mathcal{L}_{\text{SIGReg}} = \mathbb{E}_{\mathbf{v}} \left[ T_{\text{EP}} \left( \{z_c^{\top} \mathbf{v}\} \right) \right]$$

- **Stage 1 is not MedVAE**: same family, but smaller (21M),  $\times 8$  per side, trained *from scratch* on ARCADE
- **Earlier issues**: test images in the pre-training; the “25 dimensions for 90 % of the variance” measured `channel_proj`, a layer *never trained*
- Now: retrained without the test set, **3 pre-trainings**; evaluated on the layer JEPA trains (features before `conv_out`), 300 test images; probe lr chosen on validation

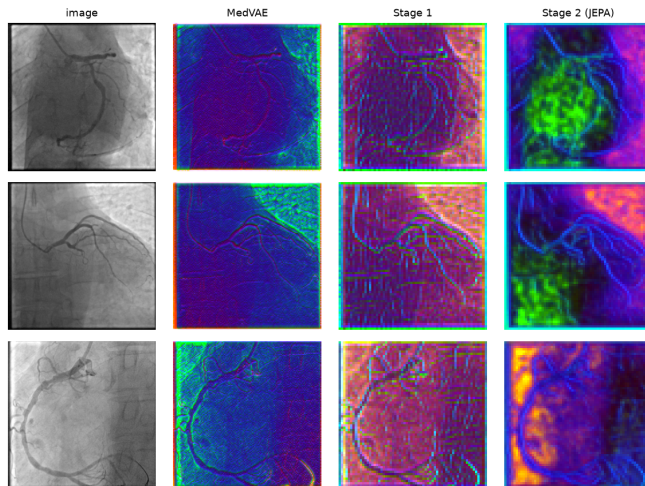
|                                     | MedVAE   | Stage 1      | Stage 2 JEPA |
|-------------------------------------|----------|--------------|--------------|
| PSNR (dB)                           | 34.8     | 31.1         | 18.6         |
| Vessel linear probe (Dice)          | 0.517    | <b>0.561</b> | 0.536        |
| Effective rank / dims for 90 % var. | 251 / 61 | 147 / 46     | 156 / 50     |

# Results — reconstruction and features



- **Frozen** decoder: the adapted encoder is no longer decodable (18.6 dB)
- JEPA **lowers** vessel separability vs stage 1 ( $-0.025$ , 3 pre-trainings out of 3), ends level with MedVAE; **no contraction** (rank 147  $\rightarrow$  156, no consistent direction)
- Best features: stage 1, before JEPA

# Results — RGB PCA of the intermediate features



Stage 2 features: large homogeneous regions, vessels still visible.

## Integration for multi-class segmentation

**Task.** Fine segmentation of coronary arteries on ARCADE:

- 26 classes (the background and 25 segments)
- thin, tubular, low-contrast structures
- extreme imbalance ( $\approx 95\%$  background)

**Tool.** MedVAE, a pre-trained medical variational autoencoder:  $\times 16$  compression in number of values ( $512^2 \rightarrow 128^2 \times 1$ ), but **lossy**.

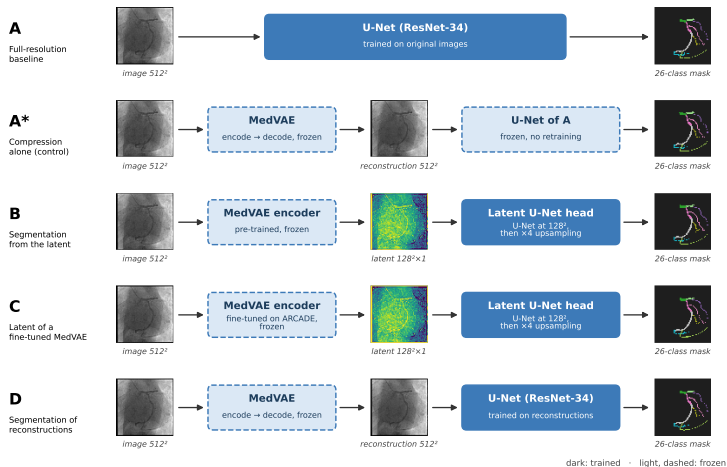
## Two questions

- 1 Does the compression preserve enough anatomical information for segmentation?
- 2 Where to integrate it: **latent** space or **pixel** space?

# Method: five conditions and two controls

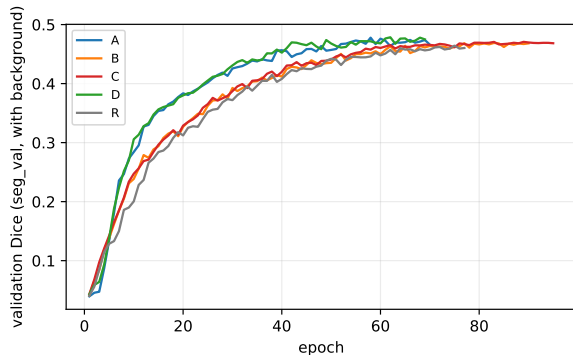
- **A** — full-resolution U-Net
- **A\*** — U-Net of A on reconstructions, **no retraining**
- **B** — U-Net head at the latent resolution, frozen MedVAE encoder
- **C** — like B, MedVAE fine-tuned on ARCADE
- **D** — U-Net retrained on reconstructed images
- **R, A<sup>↓↑</sup>** — controls:  $4 \times 4$  average pooling instead of MedVAE

3 seeds per condition; artery Dice (background excluded) on the 300 test images.



## Result 1: the latent works, like a downsampled image

- B: **0.411**, C: **0.410** artery Dice, i.e. **96 %** of A (0.428), from a  $16\times$  smaller input
- **But** the same head on the image downsampled to  $128 \times 128$  (R): **0.405** (B - R = +0.006, not established)
- **Earlier “the latent fails” (Dice 0.04)** was a training issue:
  - head learning rate  $5 \times 10^{-5}$ , too low  $\rightarrow 10^{-3}$
  - no convolution at the latent resolution  $\rightarrow$  small U-Net head
- Fine-tuning MedVAE (C): no effect (C - B = -0.001)

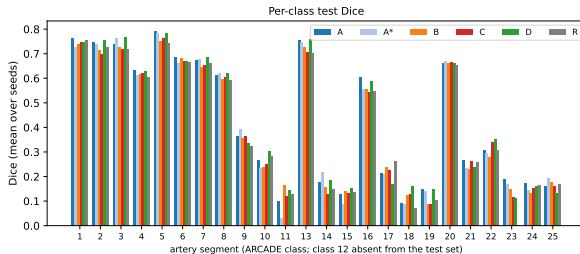
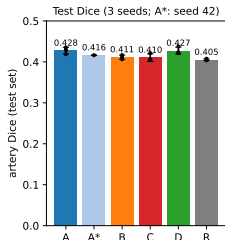


Validation Dice, mean over 3 seeds

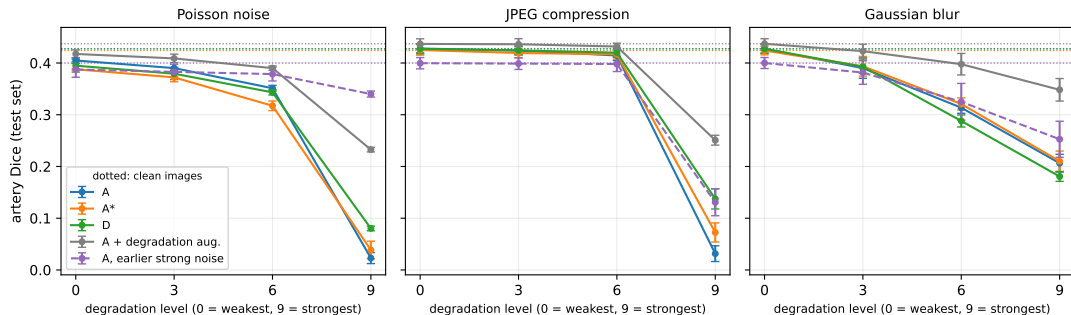
## Result 2: pixel space preserves, adaptation adds nothing

| Cond.                  | Artery Dice       | Artery IoU |
|------------------------|-------------------|------------|
| A                      | $0.428 \pm 0.008$ | 0.309      |
| A*                     | $0.424 \pm 0.008$ | 0.305      |
| B                      | $0.411 \pm 0.005$ | 0.292      |
| C                      | $0.410 \pm 0.009$ | 0.291      |
| D                      | $0.427 \pm 0.009$ | 0.309      |
| R                      | $0.405 \pm 0.002$ | 0.288      |
| A $\downarrow\uparrow$ | $0.389 \pm 0.018$ | 0.275      |

- **Compression alone costs little:**  $A^* \approx A$  ( $-0.004$ ), whereas naive downsampling costs  $-0.039$  ( $A^{\downarrow\uparrow}$ )
- **D  $\approx$  A** (0.000, within seed variability)
- **Earlier “D gains on thin segments” (+0.017):** gone with 3 seeds and the corrected augmentation

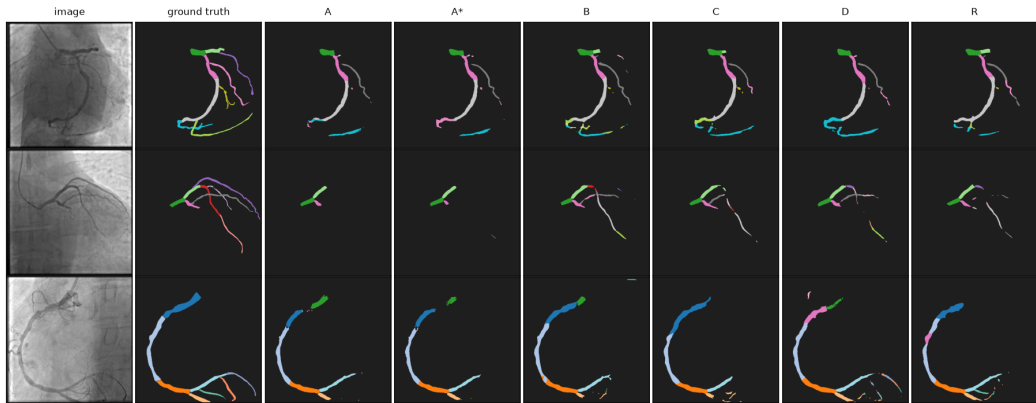


# Segmentation of degraded images



- MedVAE's noise reduction does **not** help:  $A^* \leq A$  under Poisson noise,  $D \approx A$
- Robustness comes from training: realistic degradations in A's augmentation  $\rightarrow$  strongest Poisson  $0.02 \rightarrow 0.23$ , JPEG  $0.03 \rightarrow 0.25$ , blur  $0.21 \rightarrow 0.35$ , at no cost on clean images (same degradation families as in training)

# Qualitative predictions



Seed-42 models on three test images: A, A\* and D nearly identical; B, C (latent) and R (downsampled image) follow the coronary tree at  $128 \times 128$ .

# General conclusion

Across four complementary axes, we probed the behaviour of MedVAE, a generic medical compressor, on the demanding domain of coronary angiography:

- **Robustness:** reduces Poisson noise (less than a simple filter), restores neither JPEG nor blur — and does *not* make segmentation more robust.
- **FiLM:** fine-tuning on ARCADE gains 7 dB; conditioning on a quality score adds nothing, the score being a function of the image itself.
- **JEPA:** with a frozen decoder, degrades reconstruction and vessel separability; no dimensional contraction.
- **Segmentation:**  $\times 16$  compression keeps the anatomy in pixel space ( $A^* \approx D \approx A$ , unlike naive downsampling); the latent reaches 96 % of A's Dice, but so does a downsampled image.

**Takeaway.** A faithful compression, reusable in pixel space; its latent is usable, not better than downsampling; several first-version conclusions reversed once the protocol was fixed (right reference, same training, several seeds, trained layer).

## Toward a *foundation model* for radiation-based medical imaging

Unify the explored axes into a single training objective.

- **Hybrid JEPA + reconstruction objective:** predict the masked latent patches while keeping the encoder anchored to a decodable space.
- **Physically grounded degradations:** quantum noise, contrast/brightness, multiplicative fields, anisotropic motion blur (cardiac/respiratory).
- **Conditioning (CVAE)** on DICOM metadata *not* derived from the image (modality, dose, acquisition parameters) → disentangled content / acquisition-quality representations.
- **Downstream validation:** classification, detection, multi-modality segmentation (ARCADE as a demanding test bed).